

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 20)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
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8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0	1	.	1
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Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

0	1	.	2
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Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

[illegible]

0	1	.	3
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Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

1

2

3

10

0 2 . 1

Explain the difference between in-situ and ex-situ conservation and state when ex-situ conservation is needed.

[2 marks]

0 2 . 2

Give **four** criteria used to select species for captive breeding and release programmes.

[4 marks]

1

2

3

4

0 2 . 3

Explain why each of the following species would be difficult to keep or breed in captivity.

[4 marks]

Blue whale

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Large blue butterfly

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10

0 3

The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

0 3

1

Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]

Answer

thousand tonnes

0 3

2

Calculate the maximum catch that would have kept the stock constant.

[1 mark]

Answer thousand tonnes

03 . 3

Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

[3 marks]

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03 . 4

Describe how each of the following measures obtained from commercial catches can indicate overfishing.

[4 marks]

Catch per unit fishing effort

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Mean fish size and mean age

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10

0 4

Table 1 shows the numbers of breeding pairs of two seabird species on an island before and after a rat eradication programme.

Table 1

Species	Breeding pairs before eradication	Breeding pairs 8 years after eradication
Manx shearwater	45	310
Storm petrel	12	140

0 4

1

Explain why oceanic islands have many endemic species but low overall biodiversity.

[4 marks]

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0 4

2

Use Table 1 to explain why introduced rats had such a severe effect on these birds.

[3 marks]

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0 4

3

Explain how avian malaria threatens Hawaiian birds and how climate change worsens the threat.

[2 marks]

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04 . 4

Give **one** other conservation measure for oceanic islands besides eradicating introduced species.

[1 mark]

10

05 . 1

Describe **two** case studies of biological control, one successful and one that caused problems.

[4 marks]

05 . 2

Explain how pheromone traps can be used in pest control.

[2 marks]

05 . 3

Explain why the use of antibiotics as growth promoters in livestock is a concern for human health.

[3 marks]

0 5 . 4

Give **one** way farmers can support crop pollinators.

[1 mark]

10

0 6 . 1

Explain what is meant by **biomimetics** and describe **three** examples.

[4 marks]

0 6 . 2

Explain why many plants produce chemicals that are useful as human medicines, and give **two** examples.

[3 marks]

0 6 . 3

Explain why the discovery of medicines is a strong argument for conserving whole habitats rather than individual species.

[2 marks]

0 6 . 4

Give **one** example of a wildlife species used in physiological research and the human health issue it helps to understand.

[1 mark]

10

0 7

Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

Table 1

Feature	Mixed indigenous woodland	Sitka spruce plantation
Tree species	12	1
Age structure	All ages, dead wood present	All planted in the same year
Planting density	Natural, variable	Close, 2 m spacing
Timber yield / $\text{m}^3 \text{ ha}^{-1} \text{ yr}^{-1}$	4	16
Bird species	38	9
Ground flora species	60	4

0 7 . 1

Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

[6 marks]

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0 7 . 2 Explain why commercial plantations are felled before the trees reach full size.
[2 marks]

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0 7 . 3 Explain why there is little demand today for coppiced or pollarded wood.
[2 marks]

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10

08 . 1

For each of the following forms of direct exploitation, give a named species whose population has been reduced.

[5 marks]

Food

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Fashion

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Entertainment / pets

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Furniture and ornaments

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Traditional medicines

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08 . 2

Explain why species are deliberately eradicated. Give **three** categories with examples.

[3 marks]

1

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2

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3

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08 . 3

Explain why legal protection alone rarely prevents the decline of a species exploited for traditional medicine.

[2 marks]

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10

09 . 1

Give **two** natural processes that are driven by solar energy.

[2 marks]

1

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2

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09 . 2

Explain why natural processes can take place at low temperatures.

[2 marks]

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09 . 3

Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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5

10 . 1

Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

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Extra space

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10 . **2** State what is meant by a **fed** aquaculture species.

[1 mark]

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10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

OR

1 1 . 2

Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

[illegible]

Do not write outside the box

END OF QUESTIONS